

ELECTROLYSIS OF AQUEOUS SOLUTIONS

KEY VOCABULARY

Aqueous: dissolved in water — water ions are also present.

Cathode rule: hydrogen forms unless the metal is less reactive than H.

Anode rule: oxygen forms unless a halide ion is present.

Half equation: shows electrons gained or lost at an electrode.

COMMON MISCONCEPTIONS

~~—The metal always forms at the cathode.~~

✓ Only if the metal is less reactive than hydrogen.

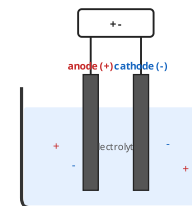
~~—Oxygen always forms at the anode.~~

✓ A halide (Cl^- , Br^-) forms the halogen instead.

~~—Water plays no part.~~

✓ Water provides H^+ and OH^- ions too.

AQUEOUS ELECTROLYSIS



1 CATHODE PRODUCT

Sodium chloride solution is electrolysed. Name the product at the **cathode**. [1 mark]

2 WHY HYDROGEN

Explain why hydrogen, not sodium, forms at the cathode. [2 marks]

3 ANODE PRODUCT

Name the product at the **anode** for sodium chloride solution. [1 mark]

4 COPPER SULFATE

Copper sulfate solution is electrolysed. Predict the products and explain why. [3 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

At the cathode you get ____ unless the metal is less reactive than it; a halide gives a ____ at the anode.

hydrogen

oxygen

metal

halogen

6 HALF EQUATION

Write the half equation for **hydrogen forming** at the cathode. [2 marks]